Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Abstract

1 Introduction

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3.1 Spanish wholesale electricity market: a sequential-market overview

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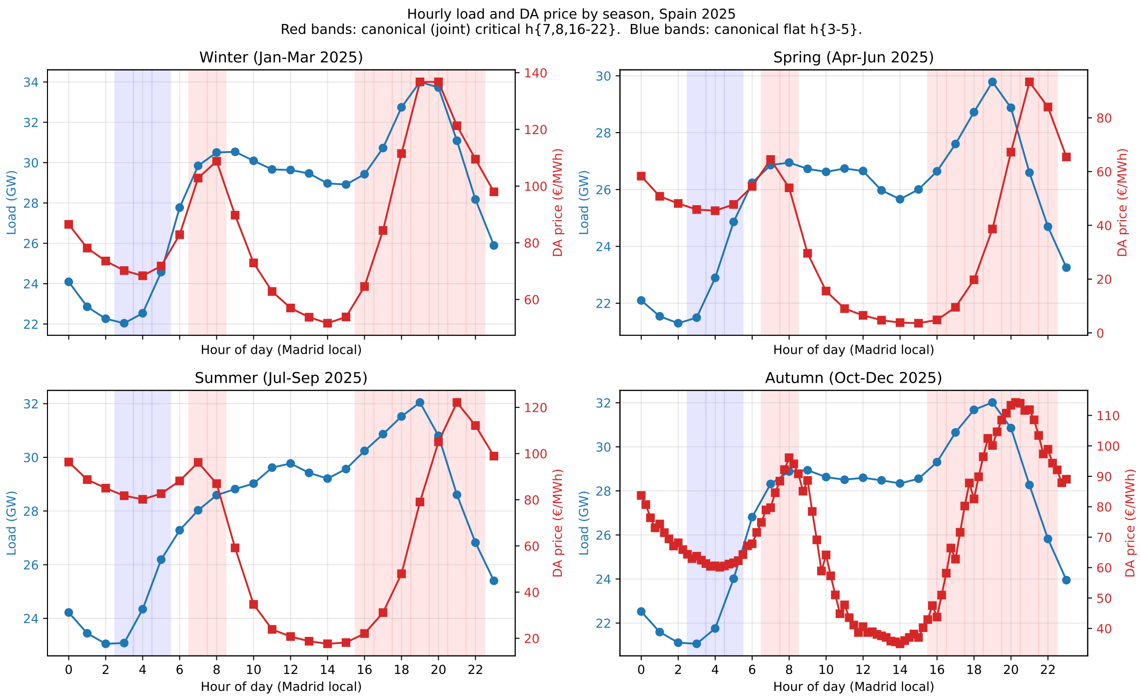

Figure 1: Hourly load and DA price profile, by season, Spain 2025. Four-panel decomposition by season. Load (left axis, blue) is realized Spanish electricity demand from ENTSO-E A65; the slope of the load curve is the demand ramp rate (visible at h6-h9 morning and h16-h20 afternoon-evening across all seasons). Red bands mark the canonical *critical hours* $h \in \{5, 6, 7, 16, 17, 18, 19\}$ (*demand surge* $\cup$ *VRE transition*). Blue bands mark the *flat hours* $h \in \{1, 2, 3\}$. The price peak h{18-22} is stable across all four seasons; spring and summer additionally show a morning ramp peak at h7-8. Flat hours are stable everywhere. The joint set is the natural cross-season-robust definition used in the headline DiD.

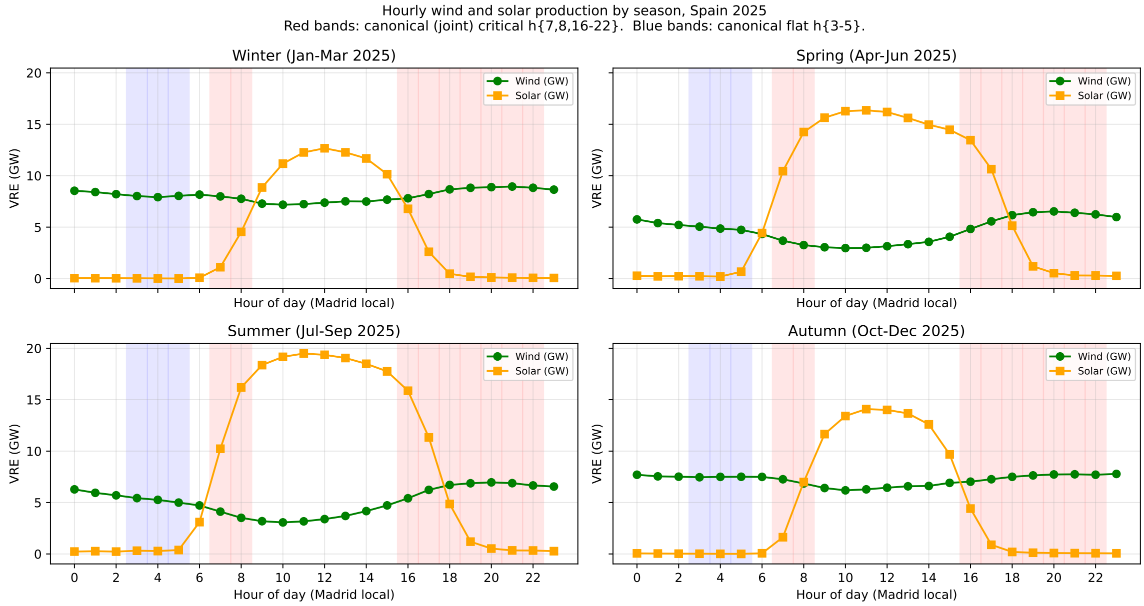

Figure 2: Hourly wind and solar production by season, Spain 2025. Four-panel decomposition by season. Red bands mark the canonical *critical hours* $h \in \{5, 6, 7, 16, 17, 18, 19\}$ (*demand surge* ∪ *VRE transition*). Blue bands mark the *flat hours* $h \in \{1, 2, 3\}$. Solar production is concentrated in midday hours and disappears in critical-hour bands; wind is broadly flat across the day with mild seasonality. The combination implies that critical hours systematically have low VRE supply year-round, supporting the use of the joint set as the strategically-relevant time-varying window.

3.5 Identification strategy and treatment-effect set

Table 1: Pivotality rate by parent firm and hour-class, CCGT only, October–December 2025. Pivotality is the unit-level price-setter rate: tranches straddle the DA clearing price (i.e., $n_{\text{below}} > 0$ and $n_{\text{above}} > 0$), evaluated per (date, unit, period) and aggregated by parent firm × hour-class. The treatment in our identification is at the *hour* level (critical vs flat). Firms enter the analysis as subjects: pivotal firms (top panel) are those expected to react to the treatment because they have a strategic margin in critical hours; non-pivotal firms (bottom panel) serve as placebo because they bid at marginal cost regardless of hour. Source: `notebooks/memos/_pivotality_by_firm_critical_hours.md`.

Firm (parent)	flat $h\{3\text{-}5\}$	critical $h\{18\text{-}22\}$	Δ crit−flat
Pivotal firms (expected to react to the hour-level treatment)			
EDP-Spain	0.72	1.00	+0.28
EDP-Portugal	0.79	0.88	+0.09
Naturgy	0.02	0.84	**+0.82**
Mid-fringe CCGT operators	0.26	0.40	+0.14
Endesa	0.12	0.31	+0.19
Iberdrola	0.004	0.25	+0.25
Non-pivotal firms (placebo)			
Engie España	0.00	0.005	0.00
Moeve (formerly Cepsa)	0.005	0.005	0.00
Repsol	0.00	0.00	0.00
TotalEnergies	0.00	0.00	0.00

3.6 Parallel-trends discussion

4 Bidding-behavior evidence of the mechanism

4.1 Per-firm bid-shape patterns at hour-of-day resolution

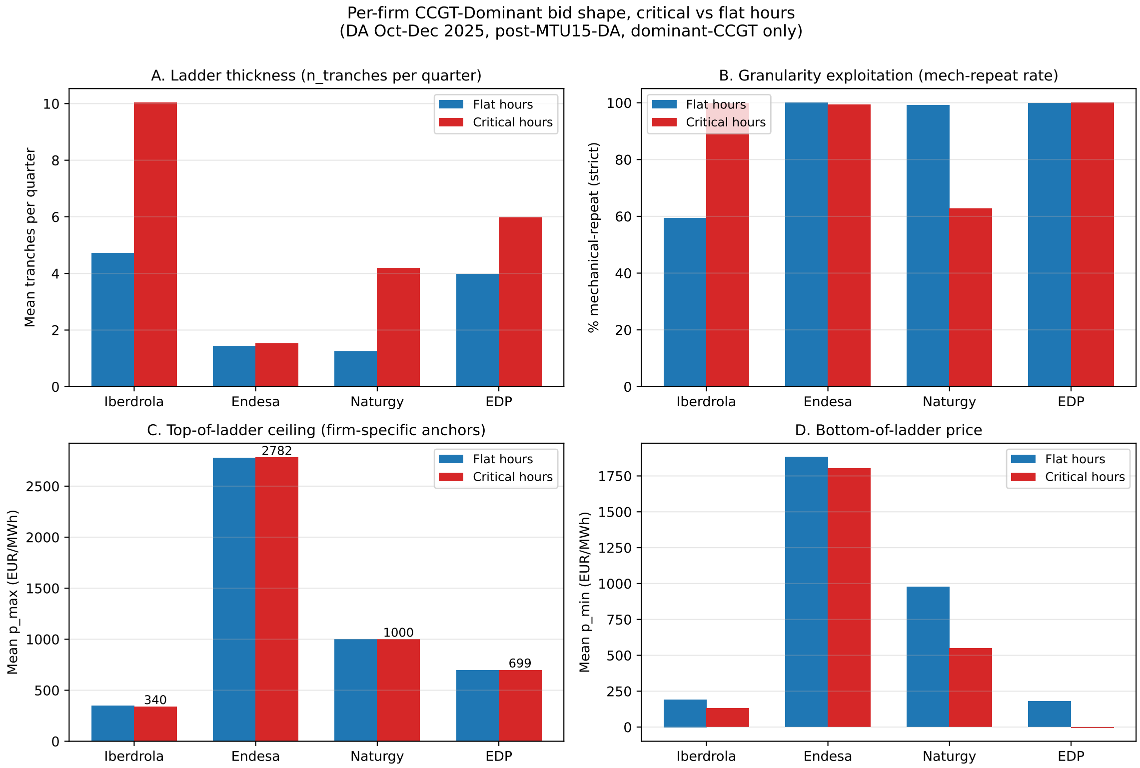

Figure 3: Per-firm CCGT bid-shape patterns, October–December 2025 day-ahead market. Four-panel decomposition of dominant CCGT bidding by firm (Iberdrola, Endesa, Naturgy, EDP-Spain). Panels show per-firm critical vs flat-hour means for tranche count, mechanical-repeat rate, top-of-ladder price, and bottom-of-ladder price. Reveals two distinct strategic types: Iberdrola (uniform enricher, richest ladders, no quarter variation) and Naturgy (granularity exploiter, varies bids quarter-by-quarter). Endesa and EDP-Spain show mid-strategic and inactive patterns respectively. Source: `scripts/analysis/firm/per_firm_hourly_ccgt_bidshape.py`.

4.2 Granularity exploitation: ladder, within-hour quarter variation, and their interaction

4.3 Pre- vs post-MTU15-DA structural changes

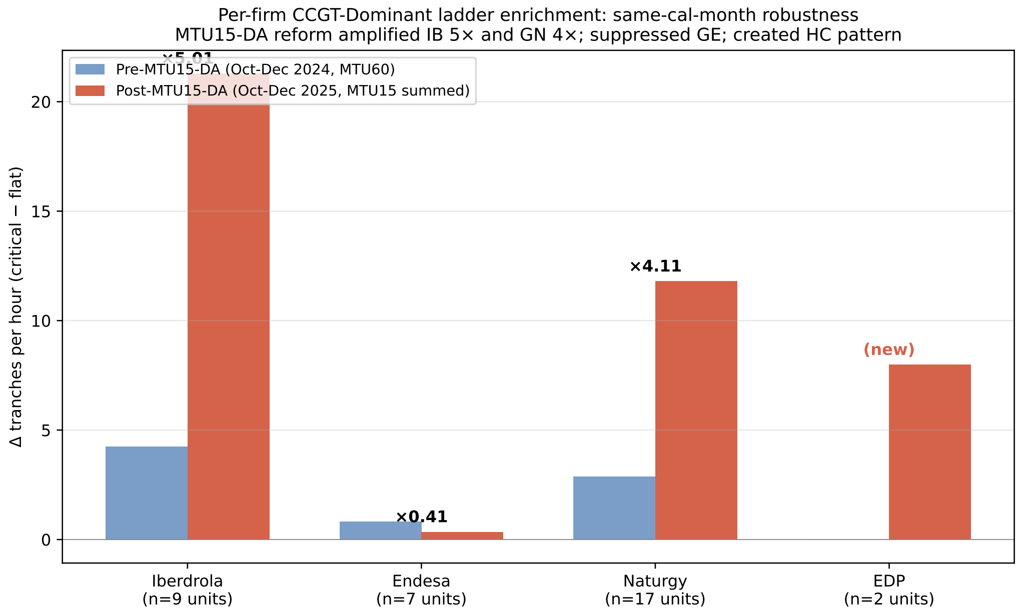

Figure 4: Pre- vs post-MTU15-DA amplification of the critical–flat differential, by firm. Same-calendar-month comparison (October–December 2024 vs October–December 2025) of CCGT bid-shape critical-flat differential by firm. Iberdrola amplifies 5.0×, Naturgy 4.1×, EDP-Spain creates a positive differential where there was none, while Endesa's pattern is suppressed. Indicates asymmetric strategic response to the granularity reform across the dominant tier. Source: `scripts/analysis/bid/per_firm_pre_vs_post_mtu15da.py`.

4.4 Heterogeneity by firm and technology

5 Main empirical results

5.1 Within-day DiD on q_2: primary outcome

Table 2: B1 – Within-day DiD on q_2 (Ito–Reguant strategic IDA upward adjustment). Outcome is q_2 in MWh per (unit, clock-hour), defined as the sum of `pibci` across IDA sessions converted to MWh and aggregated to clock-hour resolution. Specification: $q_{2,fdh} = \alpha + \beta_1 \texttt{crit}_h + \beta_2 \texttt{post}_d + \beta_3(\texttt{crit}_h \times \texttt{post}_d) + \gamma_f + \delta_{\text{DOW}(d)} + \varepsilon_{fdh}$. Firm and day-of-week fixed effects; cluster-robust standard errors by date. Sample: same-calendar-month windows, October–December 2024 and October–December 2025 (both windows post-MTU15-IDA reform of March 2025). Critical hours: $h \in \{7, 8, 16, 17, 18, 19, 20, 21, 22\}$ (joint: morning ramp $\cup$ evening peak; cross-season-robust). Flat hours: $h \in \{1, 2, 3\}$. Treatment group: firms with $\geq$ 10% pivotality in critical hours (HC, EDP-PT, GN, GE, IB). Placebo: firms with $\sim$ 0% pivotality (Repsol, Engie España, TotalEnergies, Moeve). Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; headline interpretation relies on *** given the large effective sample size.

	All firms (1)	Treatment group (2)	Placebo group (3)
β_3: `crit` $\times$ `post`	+0.01	$+4.85^{***}$	-2.79^{***}
cluster-robust SE	(0.09)	(0.98)	(0.28)
p	0.9196	0.0000	0.0000
β_1: `crit`	+0.03	−5.06	+3.03
β_2: `post`	−0.12	−14.49	+0.14
$\bar{y}$ (pre, flat)	0.12	16.51	-0.04
$\bar{y}$ (pre, crit)	0.06	11.35	3.02
$\bar{y}$ (post, flat)	0.00	1.84	0.20
$\bar{y}$ (post, crit)	-0.04	1.59	0.50
n	3,335,356	225,897	405,000
clusters G (dates)	184	184	184

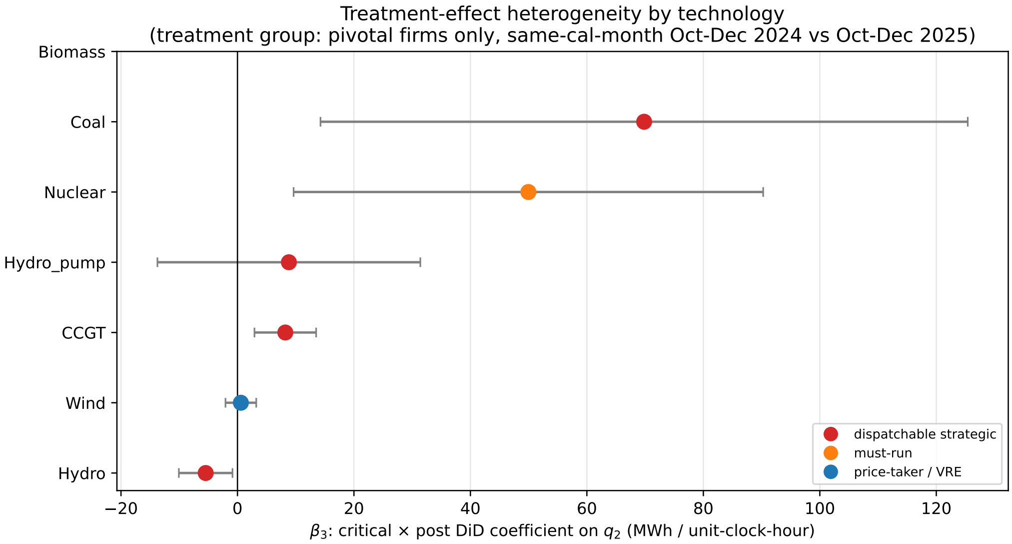

Figure 5: B1 – Treatment-effect heterogeneity by technology. Coefficient plot of β_3 (critical $\times$ post DiD interaction) on q_2, stratified by `tech_group` within the treatment group. Error bars are 95% confidence intervals using cluster-robust standard errors by date. Color coding: red = dispatchable strategic technologies (CCGT, Hydro, Hydro_pump, Coal); orange = must-run (Nuclear); blue = price-taker / VRE (Wind). The dispatchable–non-dispatchable split is the key validation of the within-day-DiD design: granularity has economic content for technologies that can strategically reshape supply, not for must-run or price-taker technologies. Source: Table 3.

Table 3: B1 – β_3 stratified by technology, treatment vs placebo group. Same specification as Table 2, restricted by technology and group. The "Predicted" column shows the model's directional prediction: + for dispatchable strategic technologies (CCGT, Coal, Hydro, Hydro_pump), 0 for must-run (Nuclear) and price-taker (Wind, Biomass). The treatment-group dispatchable techs all show large positive β_3; Wind shows the predicted null. The placebo group should show $\beta_3 \approx 0$ across all technologies. Significance markers as in Table 2.

	Treatment group		Placebo group		
Technology	β_3	SE / n	β_3	SE / n	Predicted
CCGT	+8.22**	(2.70) / 48,277	+12.84	(17.03) / 9,561	+
Hydro	−5.46*	(2.35) / 26,176	−1.90	(4.73) / 3,764	+
Hydro_pump	+8.83	(11.52) / 3,728	—	—	+
Coal	+69.83*	(28.35) / 677	—	—	+
Nuclear	+49.96*	(20.57) / 12,625	—	—	0
Wind	+0.58	(1.35) / 58,677	+0.54***	(0.16) / 220,570	0
Biomass	—	—	—	—	0

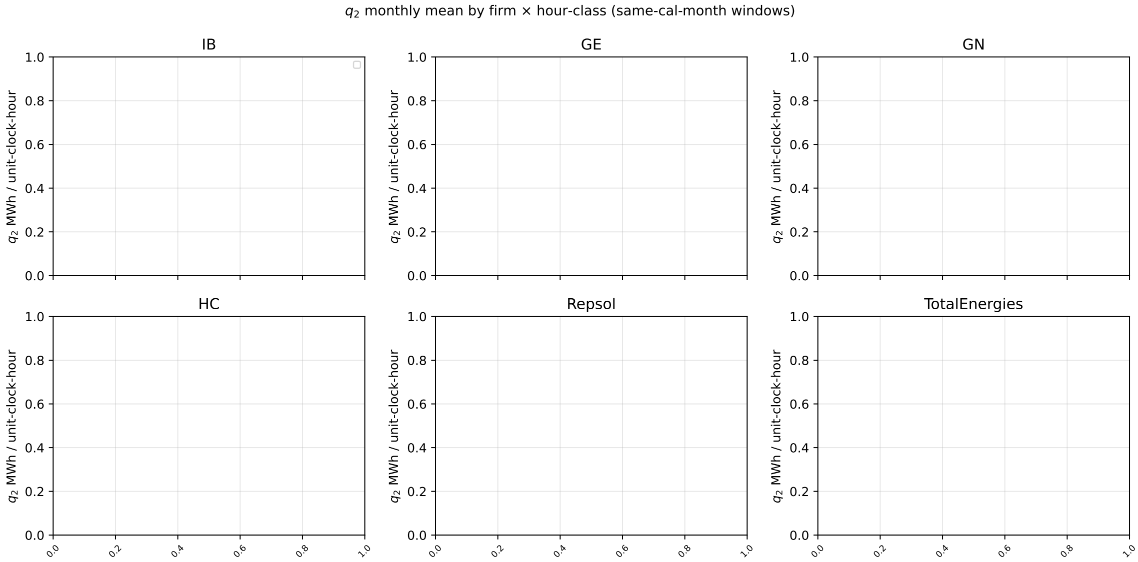

Figure 6: q_2 monthly mean by firm and hour-class, same-calendar-month windows. Six-panel monthly trajectory of q_2 (in MWh per unit-clock-hour) for the four treatment-group firms (Iberdrola, Endesa, Naturgy, EDP-Spain) and two placebo firms (Repsol, TotalEnergies). Red = critical hours $h\{18\text{-}22\}$; blue = flat hours $h\{3\text{-}5\}$. The two same-calendar-month windows (October–December 2024 and October–December 2025) bracket the MTU15-DA reform of October 1, 2025. Both windows are post-MTU15-IDA so the IDA reform of March 2025 is held constant. Source: `scripts/analysis/firm/thesis_figures.py`.

Table 4: Per-firm β_3 on q_2 (B1) and DA cleared MWh (B3). Joint per-firm view of the q_2 outcome (Ito–Reguant IDA upward adjustment) and DA cleared volume. The mechanism storyboard predicts *negative* β_3 on DA cleared (firms withhold in DA in critical hours post-reform) and *positive* β_3 on q_2 (firms add upward in IDA). EDP-PT has no `pibci` entries (Portuguese-zone units do not appear in OMIE intraday programmes); the DA-side effect is identified.

	q_2 outcome (B1)		DA cleared (B3)	
Firm	β_3	SE	β_3	SE
Iberdrola	$+9.83^{***}$	(2.79)	-24.33^{**}	(9.39)
Endesa	$+9.75^{***}$	(2.00)	$+8.12^{**}$	(2.81)
Naturgy	$+0.43$	(1.38)	-23.17^{***}	(4.91)
EDP-Spain	$+2.71$	(1.42)	$+1.39$	(2.20)
EDP-Portugal	—	—	-60.09^{***}	(8.41)

5.2 DA → IDA price wedge: supporting outcome

Table 5: B2 – Within-day DiD on the DA → IDA price wedge. Outcome is hourly Spain DA clearing price minus IDA clearing price (averaged across IDA sessions) in EUR/MWh. Same-calendar-month restriction (Oct-Dec 2024 vs Oct-Dec 2025) shows the within-day differential expanded modestly post-reform. The full-window specification (2024-01 to 2025-12) attributes most of the within-day wedge spike to the post-blackout *operación reforzada* (April–September 2025): once the reforzada × critical interaction is included, the post-MTU15-DA × critical coefficient becomes large and negative, indicating that MTU15-DA itself *reduces* the within-day wedge after controlling for the regulatory overlay. The DA-IDA wedge is therefore a noisy outcome for MTU15-DA identification absent reforzada controls.

Specification	β_3	SE	p	G
Same-cal-month (Oct-Dec 2024 vs 2025)	$+4.62^{**}$	(1.43)	0.0012	184
Full window, no controls	$+0.79$	(1.18)	0.5043	730
Full window, + reforzada × crit	-10.06^{***}	(1.39)	0.0000	730
Full window, + reforzada + MTU15-IDA × crit	-10.06^{***}	(1.39)	0.0000	730

5.3 Other outcomes (DA cleared volume, bid-shape metrics)

5.4 Conditional parallel trends with renewable and reforzada controls

Table 6: B4 – Conditional parallel trends: spec stack for q_2, treatment group only. β_3 stability under successive control additions, starting from the baseline B1 specification (firm + DOW fixed effects). Spec (2) adds Spain hourly wind and solar production levels (ENTSO-E A75); Spec (3) adds the interaction of critical-hour dummy with VRE production (allowing the VRE effect to differ across critical and flat hours); Spec (4) adds calendar-month fixed effects. The same-calendar-month design already absorbs reforzada and MTU15-IDA via the post dummy (both hold constant within each window), so no separate controls for those reforms are needed in this spec stack. Stability of β_3 across columns is the conditional parallel-trends defence of the headline result.

Specification	β_3	SE	p	G
(1) Baseline (firm + DOW FE)	$+4.85^{***}$	(0.98)	0.0000	184
(2) + wind, solar levels	$+4.63^{***}$	(0.99)	0.0000	184
(3) + crit × VRE	$+4.45^{***}$	(0.98)	0.0000	184
(4) + calendar-month FE	$+4.44^{***}$	(0.98)	0.0000	184

6 Conclusion

A Robustness checks

A.1 Sensitivity to critical-hours definition

A.2 Sensitivity to firm-partition (pivotality vs administrative)

A.3 Same-cal-month vs full-window comparisons

Table 7: B5 – Robustness panel: β_3 across specifications, treatment group on q_2. Each row reports the headline coefficient β_3 under a single sensitivity. Section B5.1 varies the critical-hours definition between four candidates (price_peak is canonical). Section B5.2 varies the firm partition between the pivotality-based set (which includes EDP-PT) and the administrative IB/GE/GN/HC tier; the two coincide because EDP-PT lacks `pibci` data so its inclusion does not alter the q_2 regression. Section B5.3/B5.5 varies the window: the full 2024–2025 panel reduces the coefficient because the pre-MTU15-DA half of the panel is post-MTU15-IDA, where q_2 magnitudes are already crushed by the IDA reform; excluding the reforzada-active months (April–September 2025) recovers a coefficient close to the same-calendar-month estimate. Section B5.4 verifies that excluding individual units (EDP-PT, ABO2G) does not change the result.

Sensitivity	β_3	SE	p	G
B5.1 — Critical-hours definition				
Canonical h{5,6,7,16-19} (demand surge ∪ VRE transition)	+4.85***	(0.98)	0.0000	184
supply_ramp h{7,8,16,17,18}	+6.20***	(1.19)	0.0000	184
price_peak h{18-22}	+8.20***	(1.49)	0.0000	184
demand_peak h{16-20}	+7.85***	(1.34)	0.0000	184
joint h{7-8, 16-22}	+6.98***	(1.27)	0.0000	184
B5.2 — Firm partition				
Pivotality-based treatment set	+4.85***	(0.98)	0.0000	184
Administrative IB/GE/GN/HC	+4.85***	(0.98)	0.0000	184
B5.3 / B5.5 — Window				
Full window 2024-2025	+1.89***	(0.30)	0.0000	730
Full window minus Apr-Sep 2025	+3.68***	(0.52)	0.0000	575
B5.4 — Sample exclusions				
Drop EDP-PT	+4.85***	(0.98)	0.0000	184
Drop ABO2G	+4.85***	(0.98)	0.0000	184
B5.6 — DST transition days (CET ↔ CEST)				
Same-cal-month, drop DST transition days	+4.96***	(0.98)	0.0000	182
Full window, drop DST transition days	+1.87***	(0.31)	0.0000	726
B5.7 — DST regime separation (clock-hour semantics differ across CEST/CET)				
Same-cal-month, CEST days only (Oct 1–25)	+0.89	(1.10)	0.4161	50
Same-cal-month, CET days only (Oct 27–Dec 31)	+6.32***	(1.22)	0.0000	132

A.4 Placebo specifications

A.5 Operational vs strategic decomposition (PIBCA vs PHF)

A.6 Bid-shape robustness in the competitive-zone restriction

B Data appendix

B.1 Sequential-market technical reference (PDBC, PDBF, PDVD, PIBCA, PHF)

B.2 File-format specifications and parser conventions

B.3 Power-vs-energy unit discipline

B.4 Coverage and missingness

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